

Another way to analyze the risk and return profile of this volatility spread opportunity is to consider first passage times. As we discussed earlier in this chapter, first passage times are random variables corresponding to the amount of time that elapses until a random variable hits a specified level.

As an example, let's say we're interested in positioning for our volatility spread to narrow, but that we're concerned about the possibility that it might widen further instead. To start, let's calculate the first passage time density to a level of, say, 7, given the current level of 12.35. This density is shown in Figure 2.23.

The modal value for this density is seven days, and the mean value is 14.9 days.

Note that the target value we've picked, 7, is slightly less than half the distance between the current value of 12.35 and the mean of 1.046. So perhaps it should be no surprise that the expected value of our first passage time density is slightly below the half-life of 15.4 days.

In general, there is no need for the half-life to correspond to the mean of the first passage time density, in part because each first passage time density is defined by a different target. For example, let's consider the first passage time density corresponding to the mean of the process, in this case, 1.046. This density is shown in Figure 2.24.

In this case, the modal value is 26 days, and the average value is 38.25 days. In comparison, the expected value of the volatility spread in the

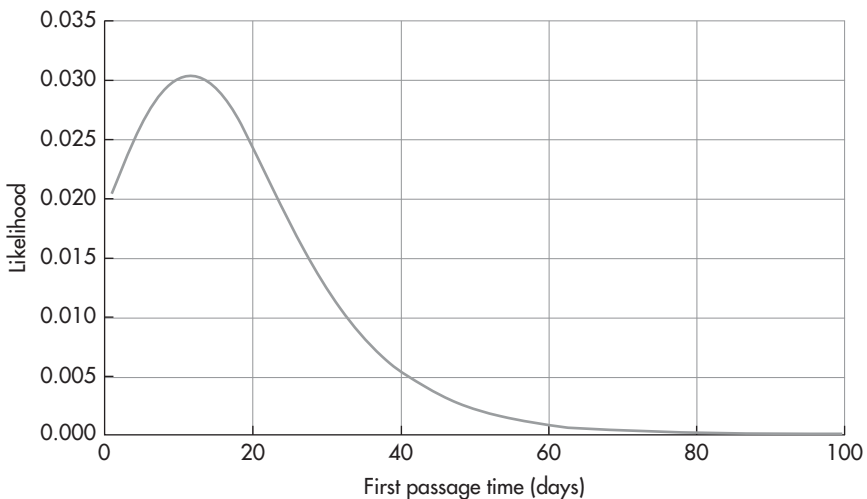


FIGURE 2.23 First passage time density of volatility spread from 12.35 to 7.

Source: Authors.

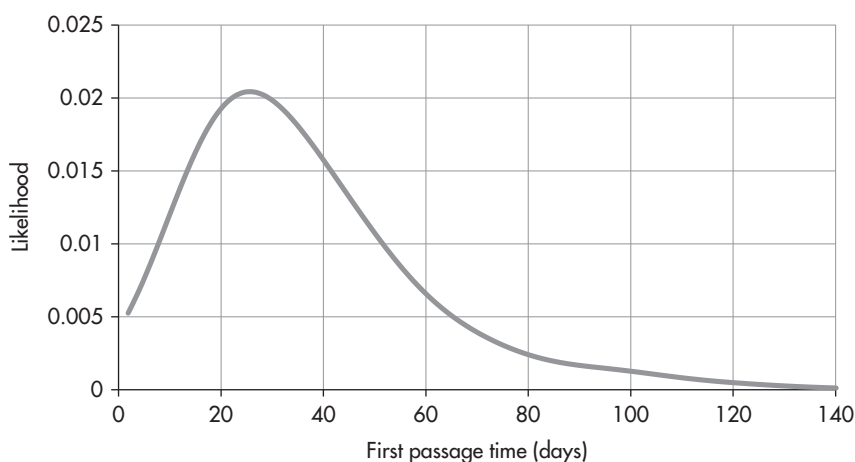


FIGURE 2.24 First passage time density of volatility spread from 12.35 to 1.046.
Source: Authors.

future, according to our model, is *never* precisely equal to its mean, since the expected value approaches the mean asymptotically from its current value. Of course, saying that an infinite amount of time must pass before the expected value equals the long-run mean isn't the same as saying that we expect an infinite amount of time to pass before the volatility spread next reaches its long-run mean.

First passage time densities are useful for a variety of reasons. For example, we can use them to make probabilistic statements about profit targets or stop levels. As an example, let's say we're interested in setting a target of 7, with the current level at 12.35. The first passage time density for this scenario is shown in Figure 2.24. We saw earlier that the average hitting time was 14.9 days, but we can make additional probabilistic statements using this density. For example, the probability of hitting our target of 7 within five days is 13.1%, and the probability of hitting the target within 10 days is 45.1%. For 15 days, the probability is 66.5%. From these statements, we also can calculate that the probability of hitting the target between five and 10 days is 32.0%.

As another example, let's say we're considering placing a stop at 16.35, four volatility points above the current level of 12.35. By calculating the first passage time density to 16.35, we can calculate that the probability of hitting our stop within 10 days is 5.2%. The probability of hitting this stop within 20 days is 8.8%. For 30 days, the probability of hitting this stop is 9.5%. It's 10.5% for 40 days, and 11.1% for 50 days. If we're uncomfortable with those

probabilities for any reason, we can calculate a new set of probabilities that correspond to another stop level.

Of course, in some of these cases, our stop would have been hit after we had taken profits on the trade, and we could use simulations to calculate multiple scenarios, such as “hitting 16.35 before hitting 8.35.”

CONCLUSION

Mean-reverting processes provide a rich set of tools for modeling financial time series exhibiting mean-reverting behavior. But care needs to be shown in choosing the appropriate tool for the job and in drawing inferences from the results.

For example, the decision to employ a nonlinear specification for the drift and/or the diffusion coefficient should be made carefully. In general, the nonlinear specification allows the model to capture a richer set of dynamics in the process, but it also allows a great opportunity to capture spurious aspects of the data that may result from nothing more than sampling error.

There are no clear rules for making these sorts of determinations, and our best advice is for the analyst to benefit from experience. We’ve attempted to provide the benefit of our experience in this chapter and throughout the book. But in the end, multiple repeated applications of the approach discussed in this chapter are likely to be of greatest use to the relative value analyst over time.

Principal Component Analysis

INTRODUCTION: GOAL AND METHOD

The market presents itself to the observer through a surface of incommensurably many data and movements. The links between those data and movements, for example, the tendency of two-year (2Y) and 10Y interest rates to rise and fall together, point toward a more or less systematic mechanism hidden in the core of the market. Our goal is to see through the surface of the market and into its structural core.

To do so, we face the problem of finding the right (degree of) formal assumptions about the structure of a market. As an extreme position, if we made no formal assumptions at all, we would remain stuck to the surface of incommensurable market data, unable to understand anything about their structural core. So, in order to reach our goal we will need to use structural terms (i.e. mathematical formalism) and hence to impose some assumptions on the market. However, we shall use those assumptions which have turned out to fit the structure of the real market well and otherwise keep them as minimal as possible, thereby leaving enough space for the market to express its own mechanisms in our form.

Principal component analysis (PCA) has only one (main) assumption: that the market is driven by a set of uncorrelated linear factors. This is not only a relatively weak assumption (allowing the market to fill in the remaining structural information, in particular about the shape and strength of each factor) but also a very useful one both for relative value (RV) analysis and hedging. It satisfies the condition of arbitrage pricing theory (see Chapter 1) and allows us to construct portfolios that are exposed to or hedged against any factor, just as specified by the investor. For the purpose of RV analysis,¹ PCA therefore appears to be a useful tool.

The main goal of this chapter is the empirical illustration of the way PCA leads us through the surface of the market into its core, where we can see its inner driving forces, gaining meaningful and deep insights into

¹Other purposes may require other tools.

market mechanisms. After developing the mathematics, we shall spend a large part of this chapter exploring the application of PCA to actual market mechanisms, thereby illustrating the way the structure of real markets reveals itself through PCA.

On one hand, the relationship between the market and mathematical form can involve problematic assumptions. On the other hand, however, it also connects the real world to its mathematical representation. PCA is therefore a link between the economy and mathematics, generating economically relevant statistics. In the example of 2Y and 10Y rates from above, a PCA could identify one factor behind moves of both rates in the same direction, and another, uncorrelated factor behind moves of both rates in opposite directions. Interpreting the identities of those factors (perhaps linked to economic data, like inflation) gives a deep understanding of the driving forces of the yield curve. In the next step, an investor who has a view on those factors (e.g. inflation) could construct through PCA the best trading position and hedge it against factors on which he has no view.

With PCA linking statistics to fundamentals, it also links the first to the second part of the book. While the focus in the preceding section was on statistics, it now shifts to statistics linked to economic insights. Correspondingly, the present goal is not an optimization of Sharpe ratios but to gain insights into the fundamental mechanisms. This will complement statistical optimality with the confidence of understanding the real economic driving forces behind a trade.

AN INTUITIVE APPROACH TO PCA

An assumption-free start to market analysis could be to simply plot data observations in a scatter chart. Figure 3.1 shows an example for 2Y versus 10Y Bund yields (from 1996 to mid-2012). Then, one can try to distinguish structural relationships. In the current example, there seems to be a strong relationship driving both yields up and down together, which has been in force more or less over the whole time period. Moreover, that relationship seems to be rather linear and can thus be approximated quite well by a straight line. In addition, there seems to be a second mechanism driving both yields in opposite directions. Compared to the first relationship, however, it appears to be weaker than the first one over the whole time period and with an impact that varies significantly over time, being quite strong when 2Y yields were around 4% and less so in other sectors. Furthermore, it is not clear whether this relationship can be reasonably assumed to be linear as well.²

²One could also remark a systemic divergence from the line for very low yield levels. The discussion following Figure 3.24 will reveal the background of that deviation.

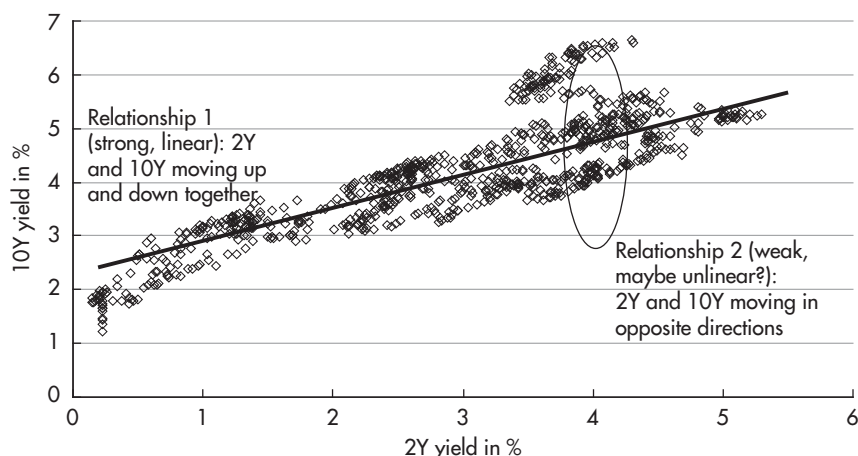


FIGURE 3.1 Structure of point cloud of 2Y and 10Y Bund yields.

Sources: data – Bloomberg; chart – Authors.

Data period: 1 Jan 1996 to 4 Jun 2012, weekly data.

Broadly speaking, PCA assumes the relationships to be linear and uncorrelated³ and provides a *quantification* of the intuitive approach from above, telling us in mathematical form:

- *which* (linear) relationships exist in a given set of market data (i.e. what the *shape* of market mechanisms looks like). In the current example, this would correspond to quantifying the impact of the relationship 1 and 2 (short and long Bund yields moving up and down together or in opposite directions) on 2Y and 10Y yields.
- *how strong* the relationships are relative to each other, that is how much of market action (yield curve variation) is explained by a particular mechanism.

In mathematical terms, the shape of market mechanisms corresponds to the eigenvectors and their relative strength to the (scaled) eigenvalues of a PCA.

³The absence of correlation is a result of the linear algebra behind a PCA. It is also a very convenient feature for the usual purposes of relative value analysis, as it allows decomposing a market into uncorrelated factors, which is the basis for analyzing them individually. However, it is also conceivable to construct a factor model with correlated factors. For example, in order to obtain a particular analytical goal it might be useful to model a market as a function of two macroeconomic variables, even if they are correlated.

Before moving on, we would like to note that a chart like Figure 3.1 can serve as a check on the suitability of the assumptions of a model for a particular market segment. In case of the assumptions of a PCA (such as linearity) seeming to violate actual market behavior, caution is advisable, and the diagnostic techniques discussed in Chapter 2 could be used.

Furthermore, this intuitive approach to PCA may give the wrong impression that PCA works like a regression, with relationship 1 being the regression line and relationship 2 representing the residuals. However, the mathematics is different, and so (usually) are the results (i.e. the relationships calculated via a PCA will in most cases differ from the regression line). We shall discuss the reasons for the difference in the section about appropriate hedging, which will also show the superiority of using PCA-based relationships for calculating hedge ratios.

FACTOR MODELS: GENERAL STRUCTURE AND DEFINITIONS

We shall now repeat the discussion from above in formal terms, thereby constructing PCA. The goal of understanding the few key mechanisms behind all market moves can be mathematically addressed by extracting the most relevant information from a given set of market data. Expressing this a bit more formally, it means reducing the dimensionality, with the remaining dimensions containing most of the information. Thus, the result of this exercise will reveal the number, strength, and shape of the market mechanisms.

We observe a number n of market data (e.g. yields) y_i^t ($i = 1, \dots, n$) at time t . The general form of a k -factor linear model is given by:

$$\begin{pmatrix} y_1^t \\ \vdots \\ y_n^t \end{pmatrix} = \sum_{i=1}^k \alpha_i^t \cdot \begin{pmatrix} f_{i1} \\ \vdots \\ f_{in} \end{pmatrix} + \begin{pmatrix} \varepsilon_1^t \\ \vdots \\ \varepsilon_n^t \end{pmatrix}$$

where α_i^t (a number which changes over time) is called the i -th factor (at time t), $\begin{pmatrix} f_{i1} \\ \vdots \\ f_{in} \end{pmatrix}$ (a vector which does not change over time) is called the i -th

factor loading and $\begin{pmatrix} \varepsilon_1^t \\ \vdots \\ \varepsilon_n^t \end{pmatrix}$ (a vector which changes over time) is called the k -factor-residual (at time t), that is the portion unexplained by the factors.

The factor loadings, which do not change over time, can be considered as containing the market mechanisms, while the factors show how much of a

specific market mechanism is active at a certain point in time. As an intuitive comparison, consider a sound mixer: the (invariable) individual sounds correspond to the factor loadings, while their (variable) strengths at a certain point in time (as adjusted by the volume regulator on the mixer board) correspond to the factors. The other way around, we can decompose the overall sound we hear into its individual components by looking at the regulators and labels on the mixer. Likewise, we are able to decompose the overall market action we observe into its individual driving forces by looking at the factors and factor loadings of a PCA.

Example: Imagine we decide to model the yield curve from 1 to 10 years

($n = 10$) by $\begin{pmatrix} y_1^t \\ \vdots \\ y_{10}^t \end{pmatrix} = \alpha_1^t \cdot \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$ (a one-factor model). This particular factor

loading allows only parallel shifts. This is an example for a very strong assumption from our side, which limits our perceptive ability to observing parallel shifts only. The reality of the market would protest against our imposed form by exhibiting large residuals.

So far, we have only assumed that market mechanisms are linear factors. The question is how much more we should assume. The answer to that question classifies factor models into two categories.

In the first category, the analyst determines the factor loadings himself. The Nelson–Siegel (NS) model is a prominent example of this approach,

assuming *a priori* that the first factor loading is the vector $\begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$ and the second

a vector whose entries follow the discount factor curve. Thus, it expands the simple one-factor model from our example above (parallel shifts) with a second factor accounting for curve steepening and flattening.

The advantage of that approach is that by choosing the factor loadings appropriately, one can ensure that the model exhibits the desired properties (e.g. is arbitrage-free). This is probably the main reason academic market analysis often prefers factor models falling into the first category.

The disadvantage is that the model results may reflect the assumptions of the analyst rather than the true market mechanisms. In the case of the NS model, for example, curve steepening and flattening moves that do not follow the shape of the assumed discount factor curve cannot be explained and will appear as residuals. Hence, the steepening mechanism of the real market will be ignored in the factor loadings (overwritten by an arbitrary *a priori* assumption) and pushed into the residuals unexplained by the factors of the model.

In the second category, the factor loadings are extracted from the market: rather than making his own *a priori* assumptions about the factor loadings (i.e. market mechanisms), the analyst lets the market reveal its own dynamics and *then a posteriori interprets* the factor loadings. For our goal of seeing into the core of the actual market (rather than imposing our hypotheses on it), the second category is thus the right place to look.

PCA is the main representative of the second category: it forces the market to reveal its mechanisms under the form of uncorrelated linear factors but usually leaves enough freedom for the market to reveal its real dynamics within that formal framework. For example, the steepening mechanism (net of direction) of the market will show up in the second factor loading just as it is, revealing the actual market dynamics rather than overwriting them with *a priori* assumptions of the analyst. Consequently, market mechanisms will be visible in the factor loadings rather than pushed into the residuals.

PCA: MATHEMATICS

Since PCA is a tool from linear algebra, we first need to represent the market in the form of a matrix. The straightforward approach is therefore to express the structural information contained in the market under consideration in the form of a covariance matrix.⁴

Technical Points

- Market data should be of sufficient length to allow the parameters of the model to be estimated with sufficient accuracy.
- Depending on the goal of the analysis, either level or change data can be used as inputs.
- It is important to use the *covariance* rather than the *correlation* matrix since the difference in volatility (sensitivity) is a key element of the analysis and must not be netted out by using correlations.

Now that we have the market information in the form of a covariance *matrix*, we can extract information by applying the powerful tools of linear algebra, transforming the covariance matrix into the orthonormal basis of its eigenvectors.

⁴Obviously, we lose some statistical information in this step, in particular above the second order. However, in practice, this is seldom of relevance for the goals of relative value analysis.

Definition: If $Ax = \lambda x$ ($x \neq 0$), for a matrix A , then the vector x is called an eigenvector of A and the number λ is the associated eigenvalue of A .

Hence, an eigenvector does not change its direction, only its length, when the matrix is applied to it. For example, if A represents the rotation of a globe by 90 degrees, then the rotation axis through the two poles is an eigenvector with an associated eigenvalue of one. Another example: if one holds a ball with both hands and squeezes it, the line connecting both hands contains eigenvectors, whose associated eigenvalue is less than one.

Theorem: For every positive semi-definite covariance matrix⁵ Cov , it is true that

$$Cov = B^{-1} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_n \end{pmatrix} B,$$

where λ_i are the eigenvalues of Cov and the columns of B consist of the eigenvectors of Cov .

Intuitive interpretation: The matrix B acts like a transformation of the coordinate system and allows us to consider the covariance matrix from the perspective of the orthonormal basis given by its eigenvectors. Since these eigenvectors are orthogonal, they decompose the covariance matrix into uncorrelated relationships. Moreover, the eigenvector associated with the greatest absolute eigenvalue points to the direction of the highest variation, that is, it represents the most important structural relationship in the market.

Example: $Cov = \begin{pmatrix} 7 & 3 \\ 3 & 2 \end{pmatrix}$ has the eigenvalues 8.4 and 0.6 with eigenvectors (e.g.) $\begin{pmatrix} 0.91 \\ 0.42 \end{pmatrix}$ and $\begin{pmatrix} -0.42 \\ 0.91 \end{pmatrix}$. Figure 3.2 shows the image of the unit circle under Cov and illustrates that most variation occurs along the direction of the first eigenvector.

⁵In fact, this theorem is true for every matrix that is symmetric and positive semi-definite.